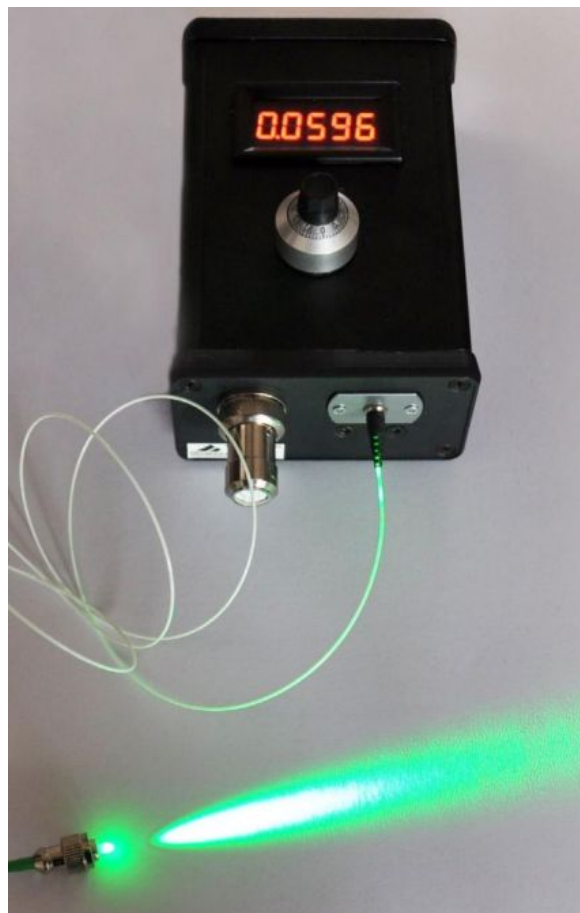


DSC

Compact manual laser diode controller with an integrated fiber-coupled laser diode. Designed for laboratory, prototyping, and field applications where simplicity, portability, and safety are critical.

MAIN FEATURES

- Compatible with LD4B series fiber-coupled laser diodes. Wavelength range: 500–1650 nm
- Constant current operation. Laser current is set via rotary knob and displayed on a digital screen
- Max output current: up to 250 mA, limited internally for the specific laser diode installed
- USB Type-C power input. Operates from any 5 V / 1 A USB power source (e.g. charger, power bank). Input voltage range: 4.5 to 5.5 V. Maximum current consumption: 0.8 A
- Built-in safety protections
 - Overcurrent protection
 - Overtemperature protection
 - ESD protection
 - Prevents enabling without current limiter
- Integrated active cooling. Internal fan cooling—no external heatsink required
- Compact enclosure. Dimensions: 78 × 120 × 43 mm
- Manufactured in Poland by [Semicon](https://semicon.com.pl)



Version 25.1

DSC

OPERATION MANUAL

1. Pre-Startup Checks

- Ensure that the laser fiber output is not directed toward the eyes or reflective surfaces.
- Verify that the power switch is in the OFF position.
- Rotate the current adjustment knob fully counterclockwise to set the current to the minimum value.

2. Power Connection

- Connect a 5 V USB Type-C power source using a suitable cable.
- Confirm that the power source and cable are capable of supplying at least 1 A of current.

3. Startup Procedure

- Turn the power switch ON.
- Slowly turn the current adjustment knob clockwise to set the desired current.
- Refer to the laser diode specification or test datasheet to determine the appropriate operating current.

4. Shutdown Procedure

- Gradually turn the current adjustment knob fully counterclockwise to reduce the current to the minimum.
- Switch the power OFF.
- Disconnect the power cable from the device.

SAFETY INSTRUCTIONS

General Electrical Safety

- The driver operates at low voltage (5 V DC), but incorrect handling may damage the module or connected laser diode.
- Never operate the module without the integrated current limiter. Do not remove or bypass this protection circuit.

Mechanical and Environmental Handling

- Do not expose the module to mechanical shock (e.g. dropping or crushing).
- Avoid storage or operation in environments where the module may be exposed to: water or high humidity, organic solvents, aggressive acids or bases, corrosive or explosive gases, dust, salt spray, or other harsh industrial conditions.
- Disposal: Treat the module as industrial electronic waste and dispose of it according to local regulations.

Laser Safety

- This device contains a fiber-coupled laser diode. Laser emission is invisible or weakly visible but potentially hazardous.
- Never look directly into the fiber connector or emitted beam.
- Always wear laser safety eyewear appropriate for the wavelength used.
- Ensure the fiber connector is directed away from users and clearly marked as a laser hazard.

Optical Fiber Handling

- Do not bend the optical fiber with a radius smaller than 25 mm to avoid permanent damage.
- Keep the fiber connector clean and protected when not in use.
- If cleaning is needed, use a lint-free cloth and pure isopropyl alcohol. Ensure the connector is dry before reconnecting.

Thermal Management

- Do not block the integrated cooling fan during operation.
- Ensure adequate ventilation around the module to support stable operation and extend laser diode lifetime.

DSC

Characteristics, data, materials and structures specified in this datasheet are subject to change without notice. Please refer to the latest specification before use of the products.

Conflict Minerals Policy Statement

LD4B achieves business objectives and customer needs with social responsibility. We do not support or contribute to the violence and human rights violations associated with the mining of conflict minerals coming from Conflict Regions according to US "Dodd-Frank Act". When possible, our suppliers' conflict mineral statements are reviewed. We do not directly purchase Conflict Minerals from any source and do not knowingly procure any parts and products containing Conflict Minerals from Conflict Regions.

RoHS Compliance Statement

Restriction of Hazardous Substances (RoHS) directive (Directive 2011/65/EC amended with Directive (EU) 2015/863) is the directive aimed at reducing the harmful environmental impact of waste electrical equipment by restricting the use of known dangerous substances. Based on information received from our supply sources, LD4B hereby states that the banned substances listed in the RoHS directive are not found in the parts and materials used above the threshold level listed other than exceptions approved by the European Commission.

REACH Compliance Statement

Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH) is a European Union regulation 1907/2006/EC that addresses the production and use of chemical substances, and their potential impacts on human health and the environment. Based on information received from our supply sources, LD4B hereby states compliance of the parts and materials used in manufacturing to REACH regulation. LD4B does not manufacture or import any substances or preparations as defined under REACH.